AI/ML Internship – Task 3

Model Validation, Overfitting Analysis & Hyperparameter Tuning for House Price Prediction

1. Introduction

The objective of this task was to improve the performance and reliability of a House Price Prediction model using the California Housing Dataset. Unlike previous tasks that focused primarily on model development, this task emphasized model validation, overfitting detection, hyperparameter tuning, and selecting a model that generalizes well to unseen data.

The project involved evaluating multiple machine learning models, applying cross-validation techniques, tuning model parameters using GridSearchCV, and analyzing model performance using RMSE and R^2 Score. The ultimate goal was to build a robust and reliable prediction model while minimizing overfitting.

2. Overfitting Analysis

Overfitting occurs when a machine learning model learns the training data too well, including noise and unnecessary patterns, resulting in poor performance on unseen data. To analyze overfitting, the Decision Tree Regressor was evaluated on both training and testing datasets before and after hyperparameter tuning.

Before Hyperparameter Tuning

Metric	Training Data	Testing Data
RMSE	0.6959	0.7242
R^2 Score	0.6377	0.5997

The baseline Decision Tree model showed relatively similar performance on both training and testing datasets. The difference between training and testing metrics was small, indicating that severe overfitting was not present. However, the model's predictive performance was limited, suggesting that there was room for improvement through parameter optimization.

After Hyperparameter Tuning

Metric	Training Data	Testing Data
RMSE	0.4866	0.6389
R^2 Score	0.8229	0.6885

After tuning, both RMSE and R^2 Score improved significantly. The model achieved a much lower prediction error and explained a larger proportion of variance in house prices. Although the gap between training and testing performance increased slightly, the substantial improvement in testing performance indicates that the model learned more meaningful patterns from the data.

The results demonstrate that the tuned model achieved better predictive capability while maintaining acceptable generalization performance.

3. Hyperparameter Tuning Approach

To improve model performance, hyperparameter tuning was performed using GridSearchCV. GridSearchCV systematically evaluates multiple combinations of hyperparameters using cross-validation and identifies the configuration that produces the best performance.

The Decision Tree Regressor was selected for tuning because it showed the strongest performance among the baseline models in Task 2.

The following parameters were optimized:

- Maximum Tree Depth (max_depth)
- Minimum Samples Required for Split (min_samples_split)
- Minimum Samples Required at Leaf Node (min_samples_leaf)

A 5-Fold Cross Validation strategy was used during tuning. In this approach, the dataset is divided into five subsets. The model is trained on four subsets and validated on the remaining subset. This process is repeated five times, ensuring that every data point is used for validation exactly once.

Cross-validation provides a more reliable estimate of model performance compared to a single train-test split because the evaluation is performed on multiple data partitions.

The tuning process successfully identified a better-performing model configuration that reduced prediction errors and improved the model's ability to generalize to unseen data.

4. Final Model Justification

The Tuned Decision Tree Regressor was selected as the final model because it demonstrated the best overall balance between predictive accuracy and reliability.

Several factors influenced this decision:

Improved Prediction Performance

The tuned model achieved lower RMSE and higher R^2 Score compared to the baseline Decision Tree model and the regression models developed in Task 2. This indicates more accurate house price predictions.

Reliable Cross-Validation Results

Cross-validation was used to evaluate model stability across multiple folds of data. Consistent performance across folds increased confidence that the model would perform well on unseen datasets and was not dependent on a specific train-test split.

Controlled Overfitting

Although the training performance improved significantly after tuning, the testing performance also improved considerably. This indicates that the model learned useful relationships from the data rather than simply memorizing the training set. Hyperparameter tuning helped control model complexity and improve generalization.

Comparison with Previous Models

The Linear Regression, Ridge Regression, and baseline Decision Tree models from Task 2 served as benchmark models. After applying model validation and tuning techniques, the Tuned Decision Tree Regressor achieved superior performance, making it the most suitable choice for the house price prediction task.

Trade-off Between Simplicity and Performance

Linear Regression and Ridge Regression are simpler and more interpretable models. However, they were unable to capture complex non-linear relationships present in the housing dataset. The Decision Tree model, after tuning, provided better predictive accuracy while maintaining acceptable model complexity. Therefore, the slight increase in complexity was justified by the improvement in performance.

5. Conclusion

This task successfully demonstrated the importance of model validation, overfitting analysis, and hyperparameter tuning in machine learning workflows. The Decision Tree Regressor was analyzed before and after tuning, and its performance was improved using GridSearchCV and cross-validation techniques.

The tuned model achieved lower prediction error and higher explanatory power compared to the baseline model and previous Task 2 models. The use of cross-validation ensured that the model's performance was reliable and generalizable.

Overall, this task provided valuable hands-on experience in building industry-aligned machine learning workflows, validating model performance, controlling overfitting, and selecting the most appropriate model based on objective evaluation metrics.